

Required Components

Hardware Components

S.No	Component	Specification	Quantity
1	Arduino Uno R3	Microcontroller Board	1
2	HC-SR04	Ultrasonic Distance Sensor	1
3	16×2 LCD	Parallel LCD Display	1
4	Potentiometer	10 kΩ	1
5	Servo Motor	SG90 Micro Servo	1
6	Green LED	5 mm	1
7	Yellow LED	5 mm	1
8	Red LED	5 mm	1
9	Piezo Buzzer	5 V	1
10	Resistors	220 Ω	3
11	Breadboard	Standard	1
12	Jumper Wires	Male-to-Male	As required

Software & IDE

Software	Purpose
Arduino IDE	Writing, compiling, and uploading the Arduino program
Tinkercad Circuits	Circuit simulation and testing
Fritzing (Optional)	Circuit schematic and PCB design